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Takagi-Sugeno fuzzy fault-tolerant control for a non-linear system

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Abstract: This paper describes the application of a new fault-tolerant control scheme for non-linear systems. The approach presented takes into account the stability and design of non-linear fuzzy inference systems based on Takagi-Sugeno (T-S) fuzzy models. The paper derives the necessary conditions for the assignability of eigenvalues to a region in the s-plane and the necessary conditions to guarantee the stability of adaptive fuzzy models for both diagnostic T-S fuzzy observers and T-S fuzzy controllers are derived. The problem is solved through using Linear Matrix Inequalities (LMI). Finally, the paper demonstrates the feasibility of this system architecture through an example, which is an application to a fault-tolerant design in detecting and isolating intermittent faults in a 3-phase induction system using digital signal processing in a hardware test-rig.

Keywords: Fault-tolerant control, multiple-model approach, linear matrix inequalities, induction motor control, parameter estimation.

1. Introduction

It is sometimes necessary to control a complex system when there are faults present or when the system undergoes set-point changes (perhaps due to unknown parameter variations). To keep the function of closed-loop control, during small faults and system changes, it is necessary to generate information about these changes for a supervision scheme **[1-5]**.

It is important to be able to carry out fault detection and isolation (FDI) before faults have a drastic effect on system performance. Even system changes should be detected and isolated. Model-based estimator schemes are used to generate "residual" signals corresponding (in simple terms) to the difference between measured and estimated variables **[6]**. The residuals are processed using either deterministic (e.g. using fixed or variable thresholds) or stochastic techniques (based upon decision theory – likelihood ratio testing, etc). Once a fault/change is detected using these methods, further procedures (e.g. structured or directional residuals or probability methods) must be used to isolate each fault.

Unfortunately, non-linear models are difficult to use and there is a natural tendency to try to fall back to multiple operating point linearisation. It is attractive to use a set of linear models with each one corresponding to a different operating point of the system. The challenge is to learn how to integrate together these "multiple-model" structures, to take account of operation at an intermediary operating point (between the definition of two models).
A promising approach to model and identify a non-linear system for FDI can be based on the input-output behaviour of local linear models. The use of these linear models – even in a multiple-model framework can result in purely linear input-output behaviour (although time-varying, according to set-point changes).

The identification of local linear models using the Takagi-Sugeno strategy is solved by fuzzy clustering obtaining partitionings of data into subsets **[7]**. Although models and interpolation between contiguous models in presence of noise can be identified using fuzzy clustering with an Auto Regressive eXogenous (ARX) approach, this has the disadvantage that some physical knowledge of the process under observation must be required (i.e. process delay and knowledge of approximate system order)

We explore the advantages of a multiple-model structure approach. We consider this structure for both the feedback control and FDI functions under a supervision scheme. The emphasis is on robust detection and isolation of incipient faults and operating point changes with prompt control re-scheduling – a form of fault-tolerant control scheme for complex non-linear systems.

The main idea is to have the following available: the process system measurements, linear models at each considered operating point and a variable (hopefully only one), which can be used as a *fuzzy inference variable* as a tool for formulating the dynamic structure used either for FDI or for control.

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The fuzzy inference variable is in fact a quantitative variable. However, qualitative properties of this variable are derived in order to combine linear models into a new non-linear dynamic structure under the multiple-model fuzzy inference modelling strategy of Takagi-Sugeno (T-S) (1985) **[8]**. In this way, the T-S model structure can be analysed **[9]** and designed to closely resemble the true non-linear input-output and state variable behaviour of the real process.

The T-S multiple-model scheme is used both for the feedback control **[10-13]** and for the design of observers for FDI **[14]**. Both are based on the principle of *parallel distributed compensation* **[15-17]**. By duality, the FDI structure is also based upon the T-S multiple-model system. This can be viewed as either a non-linear observer or as a (in a restricted sense) fuzzy observer (because of the use of the fuzzy inference variable).

For the observer design, the number of models selected depends on the region where the eigenvalues should be assigned. It should be emphasised that the number of models used is the one that guarantees the observer stability /pole assignment conditions, which depends on the LMI region selected.

Two important issues are also considered. The number of models to be used must be determined. The (T-S) fuzzy models for non-linear dynamic systems are described by a number of locally-linearised models. In this paper, different operating points are self-selected using a convex optimisation method based on eigenvalue assignment criteria and Linear Matrix Inequalities (LMI). For both fuzzy observer and fuzzy controller. For the fuzzy controller a linear feedback robust stabilization technique is utilized, **[18,19]**. As any complex system of connected subsystems, stability becomes an important consideration. Section 2 shows how using the Lyapunov approach, stability analysis can be extended to encompass the multiple-model structure, both for state estimation (observers) and control problems. This is achieved using an eigenvalue separability property which is valid as the multiple-model observer is not used for feedback control (i.e. this is only used for FDI).

2. TAKAGI-SUGENO FUZZY MODELS

The standard Gaussian radial-basis function (*RBF*) network is functionally equivalent to a restricted Takagi and Sugeno fuzzy inference model of fuzzy under certain conditions **[20]**. In this sense, the T-S models can be treated as a special case of the generalised class of *RBF* networks, but in order to keep the underlying model structure and transparent mathematics laws, the T-S fuzzy inference model is considered. The model is described by fuzzy *IF-THEN* rules at each operating point. The i_{th} rule to describe the system is as follows.

Rule *i*: *(i = 1, 2, ..., r)*

IF *z(t)* is M_i ***Then***

$$\begin{cases} \dot{x}(t) = A_i x(t) + B_i u(t) \\ y_i(t) = C_i x(t) \end{cases} \qquad (2.1)$$

2.1 Scheduling T-S Fuzzy observers for non-linear systems

The classical model based-fault diagnosis is altered in order to incorporate a new adaptive scheme. The model-based fault diagnosis (Fig. 1.) consists of a parameter estimation block to estimate unknown measurements or because the parameters are time-varying. In addition, the new scheme incorporates an adaptive law for tuning the observer based on the plant parametric model.

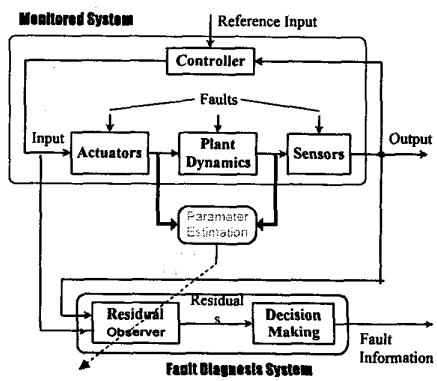

Fig. 1: Schematic diagram of an Adaptive FDI system

For a non-linear dynamic system described by the T-S fuzzy model, a fuzzy observer can be designed to estimate the system state vector. For the fuzzy observer design, it is assumed that the fuzzy system model is locally observable, i.e., all (A_x, C_x) (x = 1, ..., r) pairs are observable. Using the idea of *Parallel Distributed Compensation* (PDC) **[15,16]** for a non-linear dynamic system represented by T-S fuzzy model, a linear time-invariant observer can be associated with each rule of the fuzzy model.

Continuous System:

if $\omega(t)$ is M_x **then**

$$\left.\begin{aligned} \dot{\hat{x}}(t) &= A_x \hat{x}(t) + B_x u(t) + L_x(y(t) - \hat{y}(t)) \\ \hat{y}(t) &= C_x \hat{x}(t) \end{aligned}\right\} \qquad (2.2)$$

The overall observer dynamics will then be a weighted sum of individual linear observers.

$$\left.\begin{aligned} \dot{\hat{x}} &= \sum_{x=1}^{r} \mu_x(w)[A_x x + B_x u + L_x(y - \hat{y})] \\ \hat{y} &= \sum_{x=1}^{r} \mu_x(w) C_x \hat{x} \end{aligned}\right\} \qquad (2.2)$$

$\mu_x(w(t))$ is the grade of membership of the premise variable, *w(t)*, or the tensor product of grade of

memberships, if $w(t)$ is a vector. The membership grade function $\mu_x(w(t))$ satisfies the following constraints

$$\sum_{x=1}^{r} \mu_x(w(t)) = 1 \qquad (2.3)$$

$$0 \le \mu_x(w(t)) \le 1 \quad \forall x = 1,2,\ldots,r$$

The fuzzy observer (Eq. 2.2) has all its eigenvalues in D, if and only if there exists a symmetric matrix P such that.

1) $M^x{}_D(\bar{A}_x, P) = [\phi_{ij}P + \theta_{ij}\bar{A}_x P + \theta_{ji}P\bar{A}_x^T]_{1<i,j<k} < 0$

$$\bar{A}_x = (A_x - L_x C_x) \quad \text{for } x = 1,\cdots,r \qquad (2.4)$$

and,

2) $M^y{}_D(\bar{A}_y, P) = [\phi_{ij}P + \theta_{ij}\bar{A}_y P + \theta_{ji}P\bar{A}_y^T]_{1<i,j<k} < 0$

$$\bar{A}_y = \left(\frac{A_y - L_y C_x + A_x - L_x C_y}{2}\right) < 0$$

$$\text{for } x < y \le r \qquad (2.5)$$

Using the results presented in this section, it is possible to derive the conditions for a stable observer. More than that, it is also possible to assign the eigenvalues of the observer to a region in the s-plane. Root-clustering and eigenvalue assignment for the T-S fuzzy model using LMI region approach are described in **[14]**.

For the T-S observer, a minimum number of models must be found and the required operating points w_{r1}, w_{r2},... w_{rn} with suitable widths (Fig. 2.), so that a convex optimisation technique involving LMI has been used to find the optimum set of models (Matrices A_x, B_x, C_x D_x) used in the fuzzy observer representation, estimating the parameters of the system for each operating point.

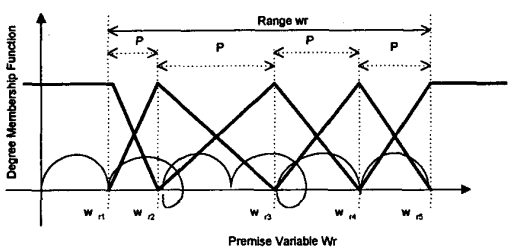

Fig. 2: Membership Fuzzy Functions

The optimisation method consists of solving the inequalities for each step. When the solution is not found in an interval, the LMI block solver is rerun a half step until it is possible to solve all the inequalities for all linear models.

2.2 Robust residual generation for discrete-time FDI

Since the aim of the robust decision-making is to minimise the false and missed-alarm rates against the effects of modelling uncertainties and unknown disturbances, an observer is proposed in order to solve the problem of complete disturbance decoupling, implying that the residual is totally unaffected by the unknown input signal, d_k, whilst it is affected by the fault f_s.

A method to produce robust residuals considering the local state space model Eq. 2.6 of the system with possible sensor faults is described in **[21]**.

$$x_k^i = A_\omega^i x_{k-1} + B_\omega^i u_k + E^i d_k \qquad (2.6)$$

$$y_k^i = Cx_{k/k} + f_{sk}$$

where $f_s \in \Re^r$ denotes sensor faults and E is the modelling error distribution matrix.

For the Takagi-Sugeno observer model, the residual r_k is obtained as:

$$r_k = y_k - C\hat{x}_k$$

$$r_k^i = y_k^i - (Cx_{k-1} - k^i Cy_k)$$

$$r^i{}_k = y_k - Cx_{k-1} + k^i Cy_k$$

with $k^i = E^i(CE^i)^{-1}$

thus, knowing the matrix E^i of disturbances k^i can be known. k^i represents the optimum eigenvalue assignment for each model in the T-S observer for the completed disturbance decoupling

3. Fuzzy T-S feedback controllers

Using the concept of *parallel distributed compensation (PDC)*, **[10,15,16]** a local linear observer is associated with each rule of the fuzzy model. The T-S Fuzzy controller can be represented by a set of ***if...then*** rules having the same premises as the fuzzy model

For continuous-time

Rule y: **if** ω_1 **is** M_{y1} ***and*** **.....** ω_2 **is** M_{y2} **then**

$$u^y(t) = K^y x(t) \qquad y = 1,2,3.....r \qquad (3.1)$$

The final output of the fuzzy controller is

$$u(t) = \frac{[\sum_{y=1}^{r} \mu_{My}(\omega(t))k^y]\ x(t)}{\sum_{y=1}^{r} \mu_{My}(\varpi(t))} = K(t)x(t) \qquad (3.2)$$

where $\mu_{My}(\omega(t))$ are the fuzzy weights involving nonlinearly the premise variables of the fuzzy model. The resulting controller is non-linear.

$\mu_{Mx}(\omega(t)$ is the grade of membership of $\omega(t)$

A sufficient condition for the stability on the feedback loop can be expressed in the following theorem.

Eigenvalue assignment: selecting the matrices L, M defining the LMI region in $D = \{z \in C: f_D(z) < 0\}$,

the different K^i are found using LMI tools, being

$$f_D(z) := L + Mz + M^T\bar{z} = [\lambda_{ij} + \mu_{ij}z + \mu_{ji}\bar{z}]_{1\le i,j\le k}$$

thus

$$M_{DI}((A_x + B_x K^x), Q) = [\lambda_{ij}Q + \mu_{ij}(A_x Q + B_x R^x)$$

$$+\mu_{ji}(A_x Q + BR^x)^T]_{1\le i,j\le m} < 0$$

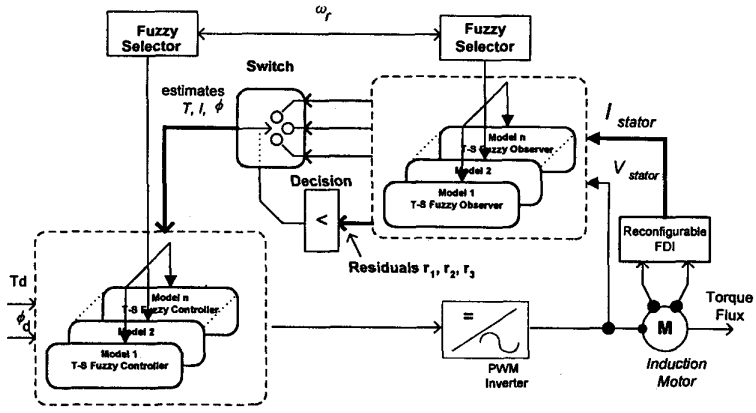

Fig. 4: Schematic diagram of the fault-tolerant control system

for x = 1,2....r (3.3)

$$M_{D2}(\{(A_x + A_y) + B_x k^y + B_y k^x)\},Q)$$

$$= \begin{bmatrix} \lambda_{ij}Q + \mu_{ij}\{(A_x + A_y)Q + B_j R^x + B_i R^x)\} + \\ \mu_{ji}\{Q(A^x + A^y)^T + (B_x R^y + R^y B_x)\}^T \end{bmatrix}_{1 \le i,j \le m} < 0$$

for x < y ≤ r (3.4)

Where $K^y = R^y Q^{-1}, y = 1,2,3...n$

The schematic diagram of both observer and controller is shown in Fig. 4, where it can be seen that a fuzzy inference engine is used to "select" the appropriate output, from those generated by the *r* parallel observers. The transition between one model to the other depends on the operating regime defined by ω.

The completed scheduling scheme is illustrated in Fig. 3. The supervision block is required not only because of the decision making but also because the design of the fuzzy observer must be achieved according to two subtasks basically: the restriction of the eigenvalues must be assigned to a specific region to ensure the system is stable in the presence of uncertainties and a fast state estimation response is also achieved with good response to the faults.

Once the parameters of the system are found the LMI solver calculates the different matrices for each linear model. The model parameters are used for the design of the adaptive fuzzy observer.

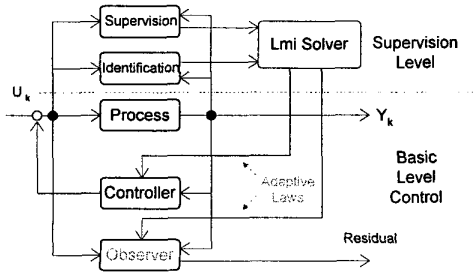

Fig. 3: Fuzzy scheduling observer scheme plus adaptive controller

4. Fault tolerant control of the induction motor in a rail way traction system: application

The performance of the proposed observer is assessed using Matlab/Simulink simulation. The FDI subsystem forms part of a fault-tolerant control system used to regulate the torque and flux in the induction motor.

The sensor fault-tolerant control scheme is run in real time using measurements taken from a hardware test-ring. The test-rig consists of a three phase AC induction motor driving a DC motor as load. The 3-phase voltages are supplied from a 100 KHz PWM-inverter. The induction motor selected is a 0.2 KW 3 phase 4 pole machine

The overall system, shown in Fig. 4, consists of a PWM inverter, which provides the necessary power to drive the machine. Since the torque and flux cannot be measured they are estimated using a bilinear observer and fed to a T-S Fuzzy Controller. The following *faults* due to gain and bias changes, noise and interference pick up, signal processing effects and *disconnections* due to continuous and intermittent loss of measurement loss of can occur in the system. A FDI/Reconfiguration block, based on fuzzy observers, performs the detection and isolation of faults and takes the necessary action to maintain the controller performance. The T-S Fuzzy representation of the induction motor observer is a class of non-linear systems. The dynamics of the induction motor observer are described by T-S fuzzy rules in Eq. 4.1. The induction motor can be modelled by the following bilinear differential equation:

$$\dot{x} = A(\omega)x + Bu = Ax + N\omega x + Bu \qquad (4.1)$$

$$y = Cx$$

where

$$x = [I_{ds}\ I_{qs}\ I_{dr}\ I_{qr}]^T\ ;\ y = [I_{ds}\ \ I_{dr}]^T\ ;$$

$$u = [V_{ds}\ V_{qs}\ V_{dr}\ V_{qr}]^T$$

where I and V denote the current and voltage respectively. The speed ω in Eq. 4.1 is referred to as the *bilinear input*.

4.1 Fuzzy observer design

For the state space representation matrices of an induction motor A, B, C, D with parameters Rs = 55; R_r = 30; L_s = 0.904; L_r = 0.904; L_m = 0.842, p=4 and with constraint **S (α = -400, r =700 ,θ =0.7854)** the following operating points are automatically found. Five models covering the speed range 0-90 rad.s^{-1} are derived using the operating points. w_{r1} = 0, w_{r2} = 30, w_{r3} = 56.25, w_{r4} = 78.3, w_{r5} = 93.75. The Universe of Discourse of ω is divided into five intervals defined by the linguistic variables *Very Small [0 30], Small [0 30 56.25], Medium [30 56.25 78.3], High [56.25 78.3 93.75] and Very High [78.3 93.75].*

Using the eigenvalue assignment conditions derived in Section 2 the following matrices L_i of the inequalities Eq. 2.4 are obtained.

$$L_1 = \begin{bmatrix} 338.0882 & 92.2197 \\ -92.2197 & 338.0882 \\ 751.5891 & 193.4495 \\ -193.4495 & 751.5891 \end{bmatrix} \quad L_2 = \begin{bmatrix} 384.9169 & -140.7486 \\ 140.7486 & 384.9169 \\ 411.8756 & 612.9361 \\ -612.9361 & 411.8756 \end{bmatrix} \quad L_3 = \begin{bmatrix} 456.4408 & -41.3692 \\ 41.3692 & 456.4408 \\ -68.3289 & 691.3912 \\ -691.3912 & -68.3289 \end{bmatrix}$$

$$L_4 = \begin{bmatrix} 529.2747 & 65.5862 \\ -65.5862 & 529.2747 \\ -354.8506 & 563.5432 \\ -563.5432 & -354.8506 \end{bmatrix} \quad L_5 = \begin{bmatrix} -4.3248 & -100.1607 \\ -4.7360 & -104.6033 \\ 100.1607 & -4.3248 \\ 104.6033 & -4.7360 \end{bmatrix}$$

Two FDI schemes are adopted to compare results. Firstly, a Dedicated Observer Scheme DOS (Clark, 1978), where each observer is excited by one output and secondly a bank of Generalised Observers **[22]**. Each observer estimates the current in one phase of the stator from the other two sensors, and is also a residual generator for fault diagnosis. The diagnostic signal-residual is generated by the comparison of the estimated and real system outputs. In so doing, the following residual signals are generated:

$$r_a = I_{phase_a} - \hat{I}_{phase_a} \quad a = 1,2,3$$

4.2 Fuzzy Controller design

The system dynamics are captured by a set of fuzzy implications which characterise local relations in the multiple model structure, employs multiple controllers with automatic switching via fuzzy rules. Conditions 3.3 and 3.4 are non-linear constraints in LMI's with P>0 such P = Q^{-1} and $K^y = R^y Q^{-1}$.

The fuzzy controller shares the same fuzzy sets and operating point with the fuzzy system and observer.

Using the same operating points (M and L) for the design of the fuzzy observer (the same matrices L and M) it is possible to assign the eigenvalues of the system in closed-loop with the following matrices.

$$K^1 = \begin{bmatrix} -7.7018 & -24.8290 & -7.7280 & -20.1932 \\ 24.8290 & -7.7018 & 20.1932 & -7.7280 \end{bmatrix} K^2 = \begin{bmatrix} -6.7248 & -48.9455 & -6.7081 & -47.2301 \\ 48.9455 & -6.7248 & 47.2301 & -6.7081 \end{bmatrix}$$

$$K^3 = \begin{bmatrix} -5.6351 & -70.0264 & -5.9565 & -70.8317 \\ 70.0264 & -5.6351 & 70.8317 & -5.9565 \end{bmatrix} K^4 = \begin{bmatrix} -4.9275 & -88.1143 & -5.1768 & -91.1135 \\ 88.1143 & -4.9275 & 91.1135 & -5.1768 \end{bmatrix}$$

$$K^5 = \begin{bmatrix} -4.32 & -100.16 & -4.73 & -104.60 \\ 100.16 & -4.32 & 104.60 & -4.7360 \end{bmatrix}$$

4.3 Results from test rig

The results presented here show that the faults can be detected quickly and accurately. However, the results gained depend on the FDI scheme adopted (DOS or GOS) and the discretising method (the order for maping the observers eigenvalue of the s plane into the z domain).

For the same S-region considered for the T-S fuzzy observer, the performance of the overall fault-tolerant control system is assessed, intermittent faults are injected successively in each of the 3 phases, A, B, C, Table 1

Fault	Time (sec)
Sensor Ia	[2.0 3.0]
Sensor Ib	[4.0 4.5]
Sensor Ic	[6.0 7.0]

Table 1: Sensor faults

The simulation demonstrates that the fault can be easily detected and isolated using the residual generated by the fuzzy observer. The residuals of the GOS (Fig. 6) are bigger than for the DOS (Fig. 5) method for the same discretisation method (first order and Tustin approximation).

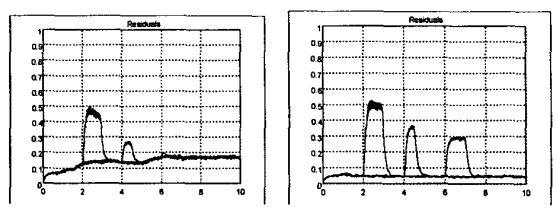

Fig. 5: Discrete DOS with first order method (left side) and Tustin's approximation (right side)

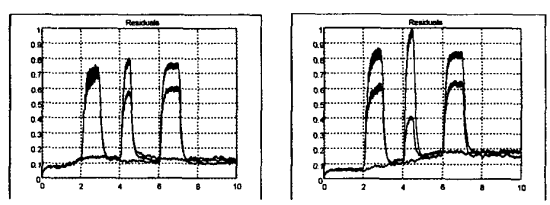

Fig. 6: Discrete GOS with first order method (left side) and Tustin's approximation (right side)

The residuals are also affected by the rotor speed ω_r. The faults at low speed will be difficult to detect with a DOS scheme, as shown in Fig.7.

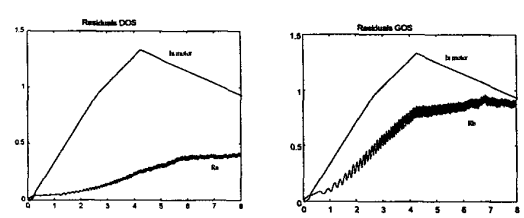

Fig. 7: DOS & GOS residuals with permanent fault

The above comparisons, Figs. from 5 to 7, of the DOS and GOS schemes using T-S fuzzy observers justify the application of the of the fuzzy observers in the fault-tolerant control Scheme shown in Fig. 4 with the configuration adopted for the isolation of the faults based on the generalised observer scheme.

5. ACKNOWLEDGEMENTS

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